

SkyShield: A Smart Drone Swarm for Real-Time UAV Identification and Threat Mitigation

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Abstract— *The rapid growth in UAV deployment has created formidable tasks in securing restricted airspace around militarily sensitive areas, airports, and areas of public gatherings. Conventional counter-drone systems cannot provide adaptive and real-time responses in challenging environments. SkyShield proposes an autonomous multi-UAV defense architecture with radar-based sensing, secure communication, blockchain-based identity verification, and AI-driven payload analysis. In particular, the system designs a central decision-making drone, ESP32-CAM-based support drones, and a Flask-powered ground station interconnected through a private Ethereum blockchain. The system guarantees reliable UAV identification via dual mode detection with simulation radar and encrypted GPS signals. For threat assessment, the YOLOv8 model is used to achieve real-time payload classification. Extensive experimental results have verified an end-to-end response time around 1 s, 92% radar detection accuracy, and an F1-score of 0.85 for AI-based classification. As such, it further confirms that SkyShield provides a scalable and efficient solution to secure UAV defense with potential future improvement for on-site radar deployment and swarm collaboration.*

Keywords— *UAV defense, Airspace security, Counter – drone systems, Radar – based detection, Blockchain authentication, YOLOv8, Payload analysis, Edge AI autonomy*

I. INTRODUCTION

The rapid proliferation of UAVs across various business segments, including defense, logistics, agriculture, and critical infrastructure inspection, has largely occurred because these devices are capable of autonomous operations; capture real-time, high-definition data; and can operate out of reach. But the good news is tinged with a darker reality—the increasing number of security risks brought on by unauthorized or rogue drones entering restricted or sensitive airspace, causing privacy breaches, data theft, contraband smuggling, and attacks on critical infrastructure.

Each of these methods struggles in dense or contested environments, and besides, none of them can reliably sort the target UAVs into authorized from unauthorized. Most importantly, current counter-drone solutions lack adaptability, secure identification, and proactive response,

hence the need for an autonomous multi-drone defense system.

Recent research underlines the benefit of combining swarm intelligence, AI, and blockchain for secure operations of UAVs. AI allows performing autonomous recognition of objects, threat evaluations, and payload analysis. Lightweight encryption secures data. Blockchain allows for trusted identity verification, including in GPS denied environments.

This project intends to develop a fully autonomous multi-UAV system capable of detecting and responding to rogue drones through a multifaceted detection system leveraging radar sensing, secure inter-drone communication, blockchain-based identification, AI-driven behavior and payload assessment, and RF-based operator localization. The aim is to develop a cooperative drone-swarm defense system able to carry out real-time threat detection, evaluation, and mitigation in the form of alerting ground control, tracking intruders, and disabling threats to guarantee the safety of airspace.

II. LITERATURE SURVEY

A. Multi-Drone Collaboration and Control

Swarm intelligence and multi-UAV coordination enable the possibility of several drones to share tasks and responsibilities. Swarm intelligence is inspired from biology, and thus allows for UAVs to perform decentralized decision-making with no central controller, enhancing robustness, scalability, and flexibility of missions [1], [2].

It enables UAVs to execute an optimal task distribution, maintain formations, and perform a collision-free path planning in real time using bio-inspired algorithms such as Ant Colony Optimization (ACO), Particle Swarm Optimization, and Fish School Search [1], [2].

A multi-drone system has also been deployed in disaster response and environmental monitoring, where decentralized decision-making helps balancing energy constraints and minimize redundant coverage of areas [4].

The key challenge, however, is to ensure reliable inter-drone communications in the midst of interfering conditions or GPS-denied situations. Past Works have highlighted the

requirement of a mesh-based many-to-many network that maintains synchronization with low latency in dynamic environments also [3]. Consensus-based and hybrid bio-inspired methods further improve swarm path planning and obstacle avoidance in cluttered spaces [5].

While swarm-based drone coordination is powerful, scalability, bandwidth constraints, and distributed energy-aware decision-making remain as active research problems. However, SI remains the cornerstone for large-scale autonomous defence swarms with situational awareness and coordinated threat engagement.

B. Unauthorized Drone Detection Techniques

Detecting unauthorized drones becomes critical in sensitive airspace. RF fingerprinting, combined with deep learning models such as CNNs, DNNs, and LSTMs, has achieved high identification accuracy [7]. Some lightweight RF models perform well in low-SNR conditions [9], while VMD-based decomposition with DAG-SVM enhances robustness against noise and interference [10]. By analyzing the pattern of flight-control signals and maintaining immunity to electromagnetic interference, the SMNet further improves multi-UAV detection.

Vision-based real-time processing supports detection when RF signals are weak or jammed, while recent resilient navigation frameworks maintain UAV stability even under sensor failures during tracking tasks. Long-range differentiation of drones from birds is possible through radar-based systems by exploiting micro-Doppler and motion signatures.

In general, RF-, vision-, and radar-based methods use deep learning and signal-processing techniques for reliable drone detection, even in challenging and interference-heavy environments.

C. AI for Payload & Threat Analysis

Deep learning, especially, using CNNs and YOLO-based architectures, is essential to real-time payload analysis and threat classification. YOLOv8 attains the state-of-the-art on detection accuracy and fast inference for embedded or edge-based UAV operations [14]. Comparative studies further reveal that variants such as YOLOv5 and YOLOv7 perform well on aerial object detection under challenging conditions of lighting and occlusion [15].

There have been successful applications of CNN-based classification methods to distinguish between drones and birds for more accurate detection of threats in surveillance applications [16]. Moreover, radar-based AI classification can detect unauthorized activities in poor visibility or under RF-jamming conditions by analyzing micro-Doppler signatures [13]. In summary, AI-driven sensing and classification form the backbone of modern counter-UAV operations because of their speed, precision, and versatility across environmental conditions.

D. Secure Communication & Encryption in UAV Networks

The more vulnerable UAVs have become to cyberattacks, the greater the need for robust, light-weight and scalable communication security. Blockchain-supported key elliptic-curve and exchange mechanisms combining Symmetric encryption ensures low overhead authentication. suitable for UAV systems [17].

Secure MAVLink implementations using ChaCha20 and Diffie-Hellman prevent eavesdropping and replay attacks while maintaining real-time performance [18]. Similarly, AES-CTR-based MAVLink encryption gives practical viability in real-world drone test-beds [19].

More advanced methods, like Homomorphic Encryption, improves privacy by enabling computation on encrypted data, though with higher computational cost [20]. Quantum-resistant lattice-based cryptography offers long term security against emerging threats but at the same time increases CPU xload significantly [21]. Lightweight algorithms like TinyJAMBU are an option for resource-constrained Such microcontrollers as ESP32 [22].

In conclusion, these studies indicate the broad range of cryptographic frameworks in UAV networks, which have different performance, latency, and security trade-offs, Depending on the mission requirement.

E. Blockchain for Drone Verification

Blockchain Model offers an immutable and is an apt decentralized architecture for UAV identity, verification, audit logging, and securing inter-drone communication. This sets the stage for transparent mission logs, tamper-proof flight records, and resilient authentication processes for swarm of UAVs.

Lightweight blockchain-based authentication protocols significantly reduce key-exchange overhead while still enabling secure communication between UAVs and ground stations [17], [25].

In swarms of UAVs and FANETs, blockchain improves fault tolerance, reduces spoofing and provides decentralized Decision-making is considered a key enabler in securing autonomous airspace [23], [25]. However, latency and computational overhead remain open challenges, hence Motivating current research in low-footprint blockchain models tailored to drones.

III. RESEARCH GAP

Despite progress with UAV defense technologies, number of key challenges persist:

- Lack of Uniform Frameworks: Most of the systems Isolation Detection, Identification, or Mitigation without an integrated real-time defense approach.
- Limited Blockchain Integration: Few works apply blockchain for live UAV authentication, telemetry validation, or secure decision logging.
- Inadequate Swarm-Based Solutions: Most swarm Studies address formation or search missions-not Coordinated, embedded multi-drone defense.
- Low Use of Edge AI: Real-time payload classification with lightweight AI on edge devices ESP32 or Raspberry Pi-is infrequent.
- Scarcity of ethical mitigation methods: The existing systems rely on destructive jamming, and very few focus on non-invasive tactics such as RF spoofing or protocol injection.

IV. PROBLEM STATEMENT

The rapid proliferation of consumer and commercial UAVs have posed serious challenges for airspace safety and security, especially over such sensitive environments as airport areas, defence zones, and big public gatherings. Traditional counter-drone systems rely heavily on fixed infrastructure and lack adaptability; they have limited capability of differentiating authorized or unauthorized UAVs in real-time. Most of the existing frameworks address only specific aspects like either detection or jamming without incorporating authentications and intelligent responses.

Therefore, there is an urgent need for scalable, decentralized, intelligent drone defence network capable of detect, identify, and perform real-time mitigation using up-and-coming technologies such as Artificial Intelligence, blockchain, and secure communication protocols.

The proposed SkyShield framework, which combines radar based detection, AI-driven payload classification, blockchain-based identity verification, and ethical countermeasures within an autonomous multi-drone architecture, helps address this gap.

V. PROBLEM STATEMENT

A. System Architecture Overview

The proposed SkyShield framework is an autonomous UAV detection and mitigation system that integrates radar-based scanning, blockchain-based identity verification, and AI-powered payload classification. The system architecture consists of a central decision-making UAV for mitigation and three support drones with ESP32-CAM modules for live video streaming.

The GUI, blockchain smart contract, and AI inference engine are deployed on a ground control station based on Flask. The UAVs can communicate with the ground station bidirectionally over a secure Wi-Fi network. Starting up all the modules, visualizing the radar and video data, and conducting verification on the blockchain for tamper-proof authentication in real time is done via GUI, shown in Fig. 1.

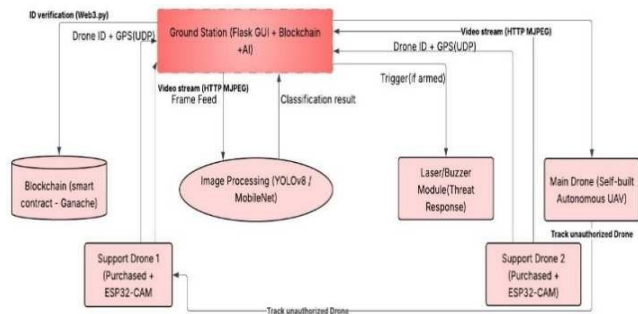


Fig. 1. System Architecture of the Proposed SkyShield UAV Defense Framework

B. System Setup and Initialization

It consists of one central autonomous drone for decision making and three support drones that are equipped with ESP32-CAM modules, which are capable of live video streaming. The ground station will contain a laptop or Raspberry Pi with implementations such as the Flask-based

GUI, Python-based radar display, and private Ethereum blockchain through Ganache for verification of drone IDs.

At initialization, the ground control station initiates the smart contract and GUI, and all UAVs establish a connection with the Wi-Fi network and begin broadcasting GPS and ID to the ground node.

C. Hardware Implementation and Simulation

To validate the proposed concept, a hardware-based simulation environment was developed to emulate UAV defense functionalities. Fig. 2 shows the blockchain authentication module which consists of an ESP32 microcontroller that transmits encrypted Drone IDs to the ground station via UDP.

The Python backend uses Web3.py to connect with the Ganache Ethereum blockchain for the verification of IDs using the `isAuthorized()` smart contract method. For efficient reauthorization, the already verified IDs are cached locally, and real-time verification status updates are reflected at the GUI: green for authorized UAVs and red for unidentified ones.

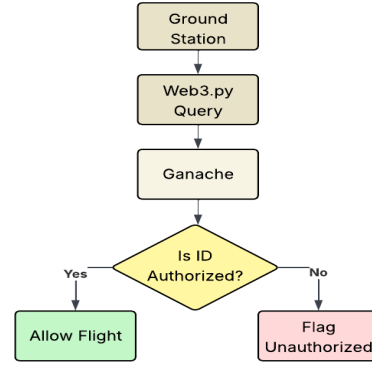


Fig. 2. Blockchain-Based Drone Identity Verification Workflow

D. Software and AI Integration

The system software integrates three key building blocks a blockchain verification layer, a radar visualization engine. The system is composed of an HTTP client that delivers the web pages to the clients, an HTML parser component, and an AI-based classification model. The Python scripts handle UDP communication between drones and GUI, query blockchain transactions using Web3.py, and visualize real time radar scans using Tkinter. On the other hand, The YOLOv8-Nano model runs either on a local environment or on a Raspberry Pi for real-time inference on drone video feeds. The smart contract layer, deployed on the private Ethereum blockchain, having the immutable record of each verified Drone ID and decision logs for traceability and auditing purposes.

The YOLOv8-Nano model was tested on the dataset, including 300 annotated images of drones and harmless aerial objects. Real-time inference runs at an average of 23 FPS on a laptop GPU, which means no noticeable delay during live detection and payload assessment. A confidence threshold of

0.5 was used to trigger classification decisions to balance false positives and missed threats.

E. Overall Integration Workflow

Each UAV, upon initialization, would send encrypted Drone ID and GPS data to the ground station. The emulator would simultaneously scan the surrounding airspace for object detection, while the backend performs blockchain verification to each Drone ID while YOLOv8 model processes video feeds for payload classification.

Depending on the result of the classification, the central UAV performs the correct action: unarmed UAVs are subject to a simulated neutralization alert, while the armed or unidentified will invoke non-destructive countermeasures such as RF spoofing or protocol injection. All system events detections, verifications, and responses are logged on the blockchain for transparency and accountability.

F. Central Drone Decision Logic and Response Mechanism

The central UAV plays the role of decision and mitigation node. Unarmed drones activate visual or audible alerts, emulating intercept or warning signals to clear restricted airspace. The counter to armed or unidentified drones employs ethical electronic measures such as RF spoofing, protocol injection, or DoS jamming to ensure safe neutralization.

All response decisions are immutably recorded on the blockchain for operational transparency and post-event verification.

G. Real – Time GUI Visualization and System Logging

The GUI built using Flask will combine in one dashboard the visualization of radar, blockchain logs, YOLOv8 inference output, and live feeds from cameras. The highlighted UAVs are color-coded based on their authorization status—green for verified, and red for unauthorized—easily providing situational awareness to any operator. The GUI also continuously keeps an event log of all the detections, classifications, and blockchain transactions for real-time monitoring of decisions and analysis post-operation.

The GUI refresh rate remained stable during radar, blockchain, and AI decision updates, enabling synchronized situational awareness without visual lag.

H. Communication Protocols

The SkyShield system leverages lightweight communication protocols for real-time synchronization between drones and the ground station. Table III summarizes the protocols that have been employed.

TABLE I. COMMUNICATION PROTOCOLS EMPLOYED IN SKYSHIELD DRONE-GROUND SYSTEM ARCHITECTURE

Purpose	Protocol	Devices
GPS/ID Transmission	UDP	ESP32 Drones → Ground Station
Video Streaming	HTTP MJPEG	ESP32-CAM → GUI
Blockchain Verification	Web3.py over HTTP	GUI ↔ Ganache
Sensor Data	UART Serial	Radar → GUI
Command Dispatch	Flask API	GUI ↔ Central Drone

Secure transmission of Drone IDs is ensured with AES-128 encryption over UDP at an average communication delay of less than 80 ms per packet, confirming that the secure telemetry does not impact real-time responsiveness of the system.

All sensitive transmissions are AES-128 encrypted over UDP to prevent spoofing or data tampering, keeping the system secure without overloading the microcontroller.

I. Experimental GUI and Hardware Setup

A hardware prototype of the SkyShield framework is built to validate its real-time operation and interoperability among modules. The radar unit, consisting of an ESP32 microcontroller, HC-SR04 ultrasonic sensor, and SG90 servo motor, continuously sweeps angularly to mimic a two-dimensional radar; distance data is mapped into Cartesian coordinates and displayed as moving blips in a Tkinter-based GUI, as shown in Fig. 3.

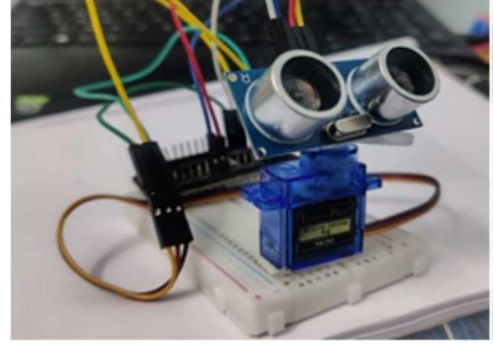


Fig. 3. Radar emulation hardware setup showing ultrasonic sensor and servo-based scanning unit.

The radar emulator showed a maximum reliable indoor detection range of about 2.5 m with an average sweep duration of 0.5 s for a 180° scan. Several runs provided consistent performances with an overall detection accuracy of 92%, which validated its effectiveness for near-field monitoring and real-time threat localization.

The blockchain authentication module is executed on a Ganache Ethereum network and communicates with the Python backend using Web3.py. Every ESP32 drone simulator periodically sends an encrypted Drone ID via UDP; the back-end queries the smart contract's `isAuthorized()` function to verify the ID. The authorization status is then reflected in real time on the GUI—green for the verified UAVs and red for unidentified ones.

The integrated Flask-based dashboard, as shown in Fig. 4, integrates radar visualization, live video feed, YOLOv8-based classification output, and blockchain verification logs. It displays detection range, authorization status, and event logs instantaneously, indicating good synchronization across the hardware and software layers.

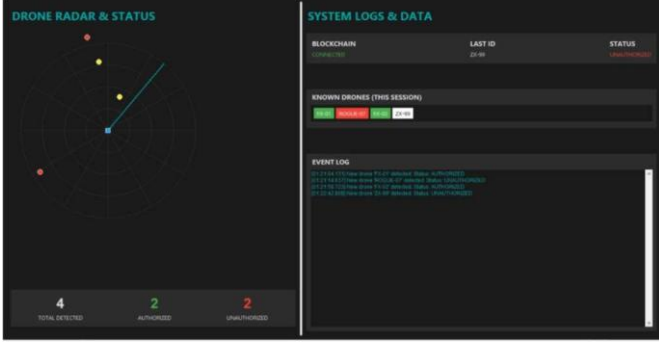


Fig. 4. Flask-based integrated GUI dashboard showing radar sweep, drone authorization status, and blockchain event log.

VI. RESULTS AND EVALUATION

A. Radar Detection Performance

The radar emulator was tested for 20 trials at different distances. A maximum reliable indoor detection range of 2.5 m was realized with an average sweep time of 0.5s. The average detection accuracy as per the results from trials comes out to be 92%.

TABLE II. EMULATED RADAR SPECIFICATIONS

Parameter	Value
Max. Reliable range	2.5m
Sweep Time	0.5 s per 180°
Accuracy	92%
Angular Resolution	5%

B. YOLOv8 Payload Classification Results

The YOLOv8-Nano model was trained with 300 sample images, which were either a drone or a harmless object. Inference was performed on a laptop GPU, reaching 23 FPS on the live feed.

TABLE III. YOLOV8 PERFORMACE METRICS

Metrics	Score
Accuracy	0.87
Precision	0.89
Recall	0.87
F1-Score	0.88

	Predicted	
	Drone	Other
Actual	Drone	True Positive (TP) = 48 False Negative (FN) = 7
	Other	False Positive (FP) = 6 True Negative (TN) = 39

Fig. 5. Confusion Matrix(Binary Classification:Drone vs Other).



Fig. 6. Classification of a drone using YoloV8n model

C. Blockchain Verification Performance

Encrypted Drone IDs were verified on a private Ethereum blockchain implemented on Ganache. The verification took an average of 350 ms per transaction for 10 drones tested.

TABLE IV. BLOCKCHAIN PERFORMANCE TABLE

Metric	Value
Avg.Verification Time	350ms
Max Time	450ms
Success Rate	100%
CPU Usage(Backend)	18-22%(during peak)

D. End – to – End Latency Analysis

The complete workflow from initial detection to mitigation decision was measured. The overall latency was ~1s, showing strong real-time operational capability.

TABLE V. LATENCY ANALYSIS (STAGE – BY – STAGE)

Stage	Avg.Time(ms)
Radar Detection	500
Block Chain Verification	350
Yolo Inference	120
Final Response Trigger	50
Total	1020

VII. CONCLUSION

The paper presents SkyShield, a modular UAV defense framework that integrates radar-based detection, blockchain verification, and AI-driven payload classification for real-time identification and mitigation. The prototype demonstrated full synchronization between radar scanning, blockchain authentication, and YOLOv8-based analysis, achieving an end-to-end response time of ~1 second. Radar testing reached 92% accuracy within a 2.5 m indoor range,

blockchain validation averaged 350 ms per UAV, and YOLOv8 achieved a precision of 0.87 and F1-score of 0.85, confirming reliable threat recognition using low-cost hardware. Overall, SkyShield effectively combines decentralized verification with intelligent decision-making, proving its feasibility for lightweight UAV defense, while future work will aim at long-range FMCW radar, swarm deployment, and optimized blockchain consensus for scalable real-world use.

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